



Test Case Specification

**Online Fact-Checking System
(Sebenarnya.my)**

Version: 1.0.0

Date: 4 January 2026

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my SEBENARNYA.MY Tidak Pasti Jangan Kongsi
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

Document Control

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Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

Version History

Version	Release Date	Section	Amendments
1.0	4 January 2026	All	Original Document

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

Table of Content

Document Control	2
Version History	3
Table of Content	4
1.0 Introduction	5
1.1 Purpose	5
1.2 References	5
2.0 Test Case	5
2.1 AI-Based News Classification	5
2.2 Multi-Language Detection & Verification	10
2.3 User Behavior Analytics & Reputation System	14

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

1.0 Introduction

1.1 Purpose

This document describes the test cases for the SeBenarnya system.

It is prepared to ensure that each functional requirement of the system is fully tested and validated.

1.2 References

This document is prepared in reference to the following documents:

1. SeBenarnya Software Requirement Specifications, SeBenarnya_SRS_1.0
2. SeBenarnya System Design Specifications, SeBenarnya_SDS_1.0
3. SeBenarnya Test Design Specifications, SeBenarnya_TDS_1.0

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

2.0 Test Case

2.1 AI-Based News Classification

Test Case ID	TC-01-001		
Related Feature ID	F005		
Objective	To verify that the system is able to classify user-submitted news content.		
Covered Test Coverage Items	TCOV-05-001, TCOV-05-002		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
User submits news content	System displays classification result (True, False, Misleading, Unverified)	User must be logged in	None

Table 2.1.1

Test Case ID	TC-01-002		
Related Feature ID	F005		
Objective	To verify that the system retrains the AI model after fact-checker verification.		
Covered Test Coverage Items	TCOV-05-003		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

Fact-checker updates classification	System updates dataset and triggers retraining	Fact-checker login required	TC-01-001
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Table 2.1.2

Test Case ID	TC-01-003		
Related Feature ID	F005		
Objective	To verify that the system detects rephrased patterns of known fake news.		
Covered Test Coverage Items	TCOV-05-004		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
Rephrased content similar to known fake news	The system detects similarity and flags it as related to known misinformation.	Requires known misinformation data	None

Table 2.1.3

Test Case ID	TC-01-004		
Related Feature ID	F005		
Objective	To verify that the system generates explainable AI results.		
Covered Test Coverage Items	TCOV-05-005		

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

Input	Expected Result	Special Procedural Requirements	Intercase Dependency
Submitted news content	System displays explanation for classification	Viewable by Admin and Fact-Checkers only	None

Table 2.1.4

Test Case ID	TC-01-005		
Related Feature ID	F005		
Objective	To verify that confidence score visibility is role-controlled.		
Covered Test Coverage Items	TCOV-05-006		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
News submitted for classification	Admin/Fact-Checker sees confidence score; Public User does not	Correct login role required	None

Table 2.1.5

Test Case ID	TC-01-006		
Related Feature ID	F005		
Objective	To verify that the AI model maintains correct classification even when exposed to adversarially manipulated news inputs.		

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

Covered Test Coverage Items	TCOV-05-007		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
News content manipulated with: • synonym replacement • spelling noise (e.g., “v1rus”, “g0vernment”) • injected irrelevant text • reordered sentence structure	AI still classifies the news correctly OR flags the content as suspicious instead of misclassifying.	Predefined adversarial dataset (fake news manipulated using NLP attack methods).	Requires TC-01-003 (rephrased detection baseline).

Table 2.1.6

Test Case ID	TC-01-007		
Related Feature ID	F005		
Objective	To confirm that the AI accurately classifies news articles with: idioms, sarcasm, passive voice, indirect claims, metaphorical language		
Covered Test Coverage Items	TCOV-05-008		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

Example: “Yeah sure, the government definitely wants everyone to get sick...” (sarcasm) Input with idioms: “This rumor has grown legs and is running wild.”	AI identifies the intended meaning, NOT the literal meaning. Correct classification OR flagged as ambiguous.	Requires contextual semantic similarity NLP model.	None
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Table 2.1.7

Test Case ID	TC-01-008		
Related Feature ID	F005		
Objective	To validate that the Explainable AI (XAI) module accurately reflects the real factors influencing the classification decision.		
Covered Test Coverage Items	TCOV-05-009		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
Submit news containing strong misinformation indicators (e.g., false medical	Explanation must include: • relevant keywords • correct semantic cues	Requires access to the model explanation APIs/logs.	Depends on TC-01-004 (XAI baseline test).

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

claims, conspiracy keywords).	known patterns used by the AI model and must NOT include irrelevant or hallucinated reasons.		
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Table 2.1.8

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

2.2 Multi-Language Detection & Verification

Test Case ID	TC-02-001		
Related Feature ID	F001 – Multi-Language Detection		
Objective	To verify that the system correctly detects the language of submitted news content (Malay, English, Mandarin).		
Covered Test Coverage Items	TCOV-02-001		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
User submits news content written purely in Malay, English, or Mandarin.	System accurately identifies and displays the correct language of the submitted content.	Language detection model enabled	None

Table 2.2.1

Test Case ID	TC-02-002		
Related Feature ID	F002 – Language Routing		
Objective	To verify that detected language content is routed to the correct language-specific classifier.		
Covered Test Coverage Items	TCOV-02-002		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

News content submitted in Malay, English, or Mandarin.	Content is routed to the corresponding language-specific AI classifier for analysis.	Language routing module configured	TC-02-001
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Table 2.2.2

Test Case ID	TC-02-003		
Related Feature ID	F003 – Cross-Language Duplicate Detection		
Objective	To verify that the system detects duplicate or semantically similar news submitted in different languages.		
Covered Test Coverage Items	TCOV-02-003		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
Same news submitted in different languages (e.g., Malay and English versions).	System identifies the submissions as duplicates or semantically similar and flags them accordingly.	Cross-language semantic comparison enabled	TC-02-002

Table 2.2.3

Test Case ID	TC-02-004
Related Feature ID	F004 – Translation Validation

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my SEBENARNYA.MY 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

Objective	To verify that translated content is validated for accuracy and meaning preservation before classification.		
Covered Test Coverage Items	TCOV-02-004		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
News content requiring translation before analysis.	Translation is validated to ensure meaning, intent, and factual context are preserved before classification proceeds.	Translation validation service enabled	TC-02-002

Table 2.2.4

Test Case ID	TC-02-005		
Related Feature ID	F006 – Multi-Language Reporting		
Objective	To verify that classification reports are generated in the user’s selected interface language.		
Covered Test Coverage Items	TCOV-02-005		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my SEBENARNYA.MY 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

User selects preferred interface language and views classification result.	System displays the classification report in the selected language with correct terminology.	Multi-language UI enabled	TC-02-002
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Table 2.2.5

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

2.3 User Behavior Analytics & Reputation System

Test Case ID	TC-03-001		
Related Feature ID	F007 – User Submission		
Objective	To verify that the system tracks and stores all user submissions linked to the correct account profile.		
Covered Test Coverage Items	REQ-03-001		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
User submits news content while logged in	Submission is stored and linked to the correct user account	User must be logged in	None

Table 2.1.14

Test Case ID	TC-03-002		
Related Feature ID	F008 – Fact-checker Verification Workflow		
Objective	To verify that the system automatically counts confirmed false submissions per user.		
Covered Test Coverage Items	REQ-03-002		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

Fact-checker verifies a submission as false	False submission count increases by one	Fact-checker must be authorized	TC-03-014
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Table 2.1.15

Test Case ID	TC-03-003		
Related Feature ID	F008 – Fact-checker Verification Workflow		
Objective	To verify that the system issues warnings or temporary suspension for repeated false submissions.		
Covered Test Coverage Items	REQ-03-003		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
User reaches predefined violation threshold	Warning is issued or account is temporarily suspended	Violation threshold must be configured	TC-03-015

Table 2.1.16

Test Case ID	TC-03-004		
Related Feature ID	F008 – Fact-checker Verification Workflow		
Objective	to verify that the system updates reputation scores correctly based on submission validity.		

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Case ID: SeBenarnya_TCS_1.0.0	

Covered Test Coverage Items	REQ-03-004		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
User has verified correct and false submissions	Reputation score increases for correct submissions and decreases for false submissions	Reputation scoring rules configured	TC-03-015

Table 2.1.17

Test Case ID	TC-03-005		
Related Feature ID	F008 – Fact-checker Verification Workflow		
Objective	To verify that automated alerts are sent when a user approaches violation thresholds.		
Covered Test Coverage Items	REQ-03-005		
Input	Expected Result	Special Procedural Requirements	Intercase Dependency
User approaches violation limit	Email or app notification is sent to the user	Notification service enabled	TC-03-015

Table 2.1.18